

Monitoring of a simulated milling circuit: Fault diagnosis and economic impact



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ABSTRACT

The early detection and root cause identification of abnormal events in industrial processes is important, to allow for timely corrective actions, ensuring continued economic operation. This paper investigates the application of statistical fault detection methods, in conjunction with process topology data-driven techniques for root cause analysis, to a simulated milling circuit. Two faults (faulty particle size analyser and rapid mill liner wear) were simulated, and the statistical monitoring techniques tested. Fault detection proved accurate, and variables closely associated with the faults were identified by the root cause analysis. The need to further formalise the selection of data for process topology generation for root cause analysis was highlighted. The milling circuit simulation and fault data has been made available as a resource for future research. Economic performance factors were developed to quantify the impact of the faults and motivate for fault detection and diagnosis.

1. Introduction

The challenge of detecting and diagnosing faults in increasingly complex industrial processes (such as the minerals industry) has received significant attention in the literature, with the focus being on data-based methods (Ding et al., 2015; Ding, 2014; Groenewald et al., 2006; Groenewald and Aldrich, 2015; Kadlec et al., 2009; Li et al., 2016; Lindner and Auret, 2015; Yang and Xiao, 2012), rather than first-principles process modelling (Agrawal et al., 2015; Guerrero et al., 2016; Légaré et al., 2016; Salazar et al., 2014; Tidiri et al., 2016; Venkatasubramanian et al., 2003a, 2003b). Here, fault detection is defined as the indication at a specific time point that a faulty condition has started occurring, or is continuing to occur. Fault diagnosis is defined here as the indication of which measured process variables are most associated with the faulty condition (also referred to as fault identification in the literature), and the subsequent classification of process variables as either symptoms or causes of the faulty condition (also referred to as root cause analysis in the literature). The goal of detecting and diagnosing faults is to reduce or prevent faulty operating conditions and ensure continued economical operation.

Principal Component Analysis (PCA) and related methods (including nonlinear and dynamic versions of PCA) are commonly used for fault detection in the process industry (Aldrich and Auret, 2013; Chioua et al., 2016; McClure and Gopaluni, 2015; Qin, 2012). Contribution-based methods are often used in conjunction with PCA-based monitoring to identify faults (Miller et al., 1998); methods include

complete, partial, diagonal, reconstruction-based and relative contributions (Alcala and Qin, 2011, 2009).

Fault diagnosis is cited as being possible by means of causality methods, which attempt to extract causal relationships between variables in the form of process topologies (Lindner and Auret, 2014; Maurya et al., 2003a, 2003b) either purely from historical data (Bauer et al., 2007; Bauer and Thornhill, 2008; de la Fuente et al., 2004; Lindner and Auret, 2014) or by making use of process knowledge (Groenewald and Aldrich, 2015; Landman and Jämsä-Jounela, 2016).

Despite the accepted economic benefits of process control in general (Bauer and Craig, 2008; Bouffard, 2015; le Roux et al., 2016; Wei and Craig, 2009) and fault detection specifically (Bin Shams et al., 2011), relatively little work has been done to illustrate the economic impact of correctly detecting and identifying faults. Extensive domain knowledge, and significant effort, can be required to develop economic performance factors, which relate process variables to economic performance. The benefit is the continuous prediction of economic performance, and the use of this prediction to assess test runs, plant modifications, control schemes, or even decide whether to shut down a plant following a fault (Olivier and Craig, 2017).

Whether detecting and diagnosing faults or assessing economic performance, any proposed method must be demonstrated or evaluated on a data set. While real operating data provides the most realistic test of any method, the “ground truth” is often missing: for example, the actual root cause of a fault may not be known. Simulated data is used in many studies, as diverse conditions can be simulated without the risk of

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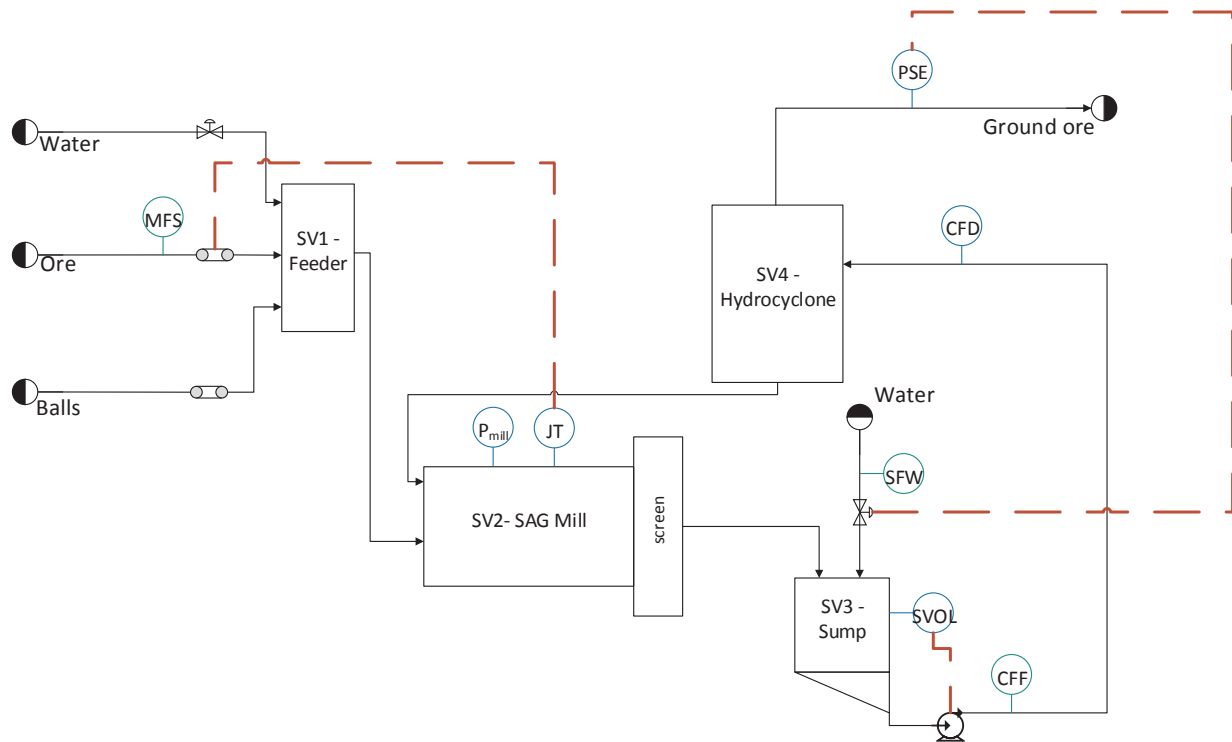


Fig. 1. Mill circuit model with control loops.

safety or financial impact. However, the simulations generating this data should be as realistic as possible, in order to provide the level of complexity and challenge required to test any proposed method, and demonstrate real-world applicability.

This paper describes the application of data-based fault detection and diagnosis techniques to a validated dynamic model of a run-of-mine milling circuit (Coetzee et al., 2010; le Roux et al., 2013), and illustrates the impact on economic performance of faulty or abnormal operation.

The proposed approach is suggested in general for more robust assessment of the actual benefit of abnormal event detection and root cause analysis in industrial processes. The dynamic simulation and simulated data are available online to promote further analysis and comparative studies.

The paper is structured as follows: Section 2 describes the milling circuit model, and the economic performance criteria that were used, and provides an overview of the data-based process monitoring approach which was applied. Section 3 shows and discusses the results of the process monitoring and fault diagnosis, and Section 4 provides conclusions and ideas for future work.

2. Methods

2.1. Dynamic mill circuit simulation with process disturbances and abnormal events

A dynamic model of a run-of-mine mill circuit has been developed to demonstrate the potential economic impact of fault detection and diagnosis. The mill circuit model and its control system is briefly described here, as well as the nature of input excitation (in the form of disturbances) to the model, the generation of normal operating conditions data and fault data. The model, which can be run in MATLAB and Simulink, has been made available as supplementary material for future research. For further details, see the Supplementary Material section of this paper.

A dynamic model (rather than industrial data) provides a well-defined data set to test the monitoring methods described below; the selected dynamic model is based on actual operations, however, and so it

provides a level of complexity that ensures that the monitoring approach is rigorously tested.

2.1.1. Mill circuit model and control

The run-of-mine grinding mill circuit model presented by le Roux et al. (2013) is considered. This simulation is based on the model found in (Coetzee et al., 2010). This is a reduced complexity nonlinear model, consisting of four defined system volumes (SV) connected as shown in Fig. 1. The system volumes are the feeder, semi-autogenous mill, sump, and hydrocyclone. The feeder is a superficial SV that exists to separate the feed to the mill circuit into the five states used in the model. Fresh feed ore, water and steel balls are added to the mill from the feeder along with underflow from the hydrocyclone. The milled ore then exits the mill and enters the sump where water is added to modify the thickness of the slurry. Lastly, a pump transports the slurry to the hydrocyclone for classification.

The variable abbreviations used in Fig. 1 are described in Table 1.

The mill circuit model was built and tested in MATLAB R2014a, R2016a and R2017a and Simulink, using the ODE45 numerical integration method.

A control system for the circuit, consisting of single loop PI controllers with input filters and output saturation (see Table 2), was implemented. The output saturation and controlled variable setpoints are based on the data presented in le Roux et al. (2013).

Control loop pairings were determined based on previous investigations into this system (Craig et al., 1992). Control loop tunings were based on Proportional-Integral controllers with the form in Eqs. (1) and (2), and using the Ciancone correlations described in Marlin (2000).

$$MV(t) = MV(0) + K_c(E(t) + \frac{1}{T_i} \int_0^t E(t') dt') \quad (1)$$

$$E(t) = SP(t) - CV(t) \quad (2)$$

The three control loops which were implemented were:

- The sump volume (SVOL) is controlled by manipulating the cyclone

Table 1
Variable abbreviations in mill circuit model.

Variable description	Variable abbreviations	Units
Mill feed solids	MFS	t/h
Mill power consumption	P _{mill}	kW
Mill charge	JT	Fraction
Sump feed water	SFW	m ³ /h
Sump volume	SVOL	m ³
Cyclone feed flow rate	CFF	m ³ /h
Cyclone feed density	CFD	t/m ³
Particle size estimate	PSE	fraction

Table 2
Control loop tuning and manipulated variable saturations.

Control loop	CV SP	K _c	T _i	MV min	MV max	Units
MV: CFF (cyclone feed flow rate)	5.99	20	0.25	100	500	m ³ /h
CV: SVOL (sump volume)						m ³
MV: MFS (mill feed solids)	0.34	42.1	9.46	0	100	t/h
CV: JT (mill charge)						fraction
MV: SFW (sump feed water)	0.67	928.6	4.54	0	400	m ³ /h
CV: PSE (particle size estimate)						fraction

feed flow rate (CFF) which manages the throughput to the hydrocyclone.

- The mill charge (JT) is controlled by manipulating the mill feed solids (MFS) which manages the overall throughput for the entire circuit as well as the electricity consumption of the mill (P_{mill}).
- The particle size estimate (PSE) is defined as the percentage of fines present in the output flow from the hydrocyclone. The PSE is controlled by manipulating the sump feed water (SFW) which indirectly manages classification in the hydrocyclone by adjusting the cyclone feed density (CFD).

The eight variables listed in Table 1 were considered as measured variables available as inputs to the statistical process monitoring method and the process topology methods.

2.1.2. Disturbance modelling and sensor noise

Two key disturbances must be rejected by the control system under normal operating conditions (NOC): feed ore size distribution variation (defined as the fraction of rocks in the feed ore, α_r); and feed ore hardness (defined as the power needed per tonne of fines produced, Φ_f).

These disturbances were modelled as random walks (Lawler and Limic, 2010). Let v_0 be the initial value of the disturbance variable v . l represents the expected gradient for this disturbance variable (i.e. mean rate of change). Parameter x is generated for each time step i as a uniform pseudorandom number with values between 0 and 1. The random walk path for variable v can then be calculated by summation:

$$v = \sum_{i=1}^N s_i \Delta t + v_0 \quad (3)$$

$$s_i = \begin{cases} +l, & x_i < 0.5 \\ -l, & x_i \geq 0.5 \end{cases} \quad (4)$$

The expected disturbance variable gradients and disturbance variable starting points were based on inspection of survey data in (Je Roux et al., 2013) and iterative tuning to achieve realistic dynamic variation.

The motivation for this disturbance modelling choice is twofold: firstly, random walks represent non-deterministic input excitation (deterministic input excitation might produce unwanted

autocorrelation artefacts). Secondly, real-world disturbance changes would not be completely time-independent: random walks enforce a certain non-independence of consecutive values.

Disturbance variation is required to recreate realistic NOC data, as well as ensuring sufficient excitation of the control system such that the expected process variable interactions can be captured.

All sensor noise was assumed to be normally distributed, with standard deviation of 1% of the measured value. The inferential sensors (JT and PSE) were modelled as having a measurement delay of 1.5 min to account for input processing. Furthermore, the control system was made more robust to noise by adding first order filters with 1 min time constants to the control loop inputs.

Disturbance variation and sensor noise were implemented for both normal operating and fault conditions.

2.1.3. Initial start-up and normal operating conditions modelling

Initial start-up of the mill circuit model involves simulation without disturbance variation or sensor noise for a simulated period of 50 h in order to reach initial steady-state. The initial start-up conditions are discarded and not considered during process monitoring.

Normal operating conditions were simulated for a total of 104 days of operation: 90 days were used as NOC training data, and 14 days for NOC test data. Process measurements are recorded every 30 s.

2.1.4. Fault modelling

Fault data were simulated for 30 days for the investigation of two different faults: a faulty particle size analyser on the product stream, and rapid wear of the mill liner. These two faults are chosen in order to provide data for both sudden and gradual faults, which must be detected and diagnosed by any process monitoring approach.

The particle size analyser sensor fault can be attributed to a faulty/dirty sensor due to dust or slurry, or simply general instrument drift. This causes the sensor to produce a bias in measurement. It is assumed that the bias would be negative: larger particles are easier to detect than smaller particles, and so the sensor is more likely to under-detect the fraction of smaller particles present, resulting in a lower PSE measurement (representing the percentage of fines in the hydrocyclone overflow). The fault was simulated as a 5% constant negative bias on the PSE reading.

The mill liner fault is a result of accelerated wear and tear to the liner, which acts as an interface between the rotating mill shell and the charge tumbling inside the mill. The liner transfers energy from the shell to the charge, and increased wear reduces the efficiency of this energy transfer, requiring more power for the same particle breakage rate, reducing the efficiency and throughput of the mill (Powell et al., 2006).

The fault is simulated by placing a bias on the calculated mill power draw. This bias is implemented only on the power draw where it is used to calculate mill breakage, and not on the mill power sensor. This is because the fault does not reduce the actual mill power draw, but does reduce the efficiency of the transfer of power to the charge.

The fault is simulated as a linear decrease in the efficiency of power transfer from mill shell to charge, from 100% at the start of the fault to 85% after 30 days. This represents abnormally rapid wear of the mill liner, possibly due to faulty material or installation. Although monitoring systems are available specifically for mill liner wear, this abnormal wear would not be detected during operation by these systems, which require mill shutdown and internal inspections.

A process monitoring approach should detect this abnormal wear (and other faults, including unknown faults) online, from process measurements (although other measurements can be included, such as vibration monitors or motor current, where available).

2.2. Economic performance criteria

The significance of abnormal events is rarely quantified in terms of

economic impact in process monitoring literature. To capture the economic impact of control and fault detection, Bin Shams et al. (2011) incorporated the variability associated with quality characteristic variables, operating costs of the process and rate of control changes which can be related to the cost of wear of actuators, in order to propose a balance between fault observability and product variability.

The milling circuit simulated in this work has been the subject of a number of studies of economic impact, most notably in comparing the economic performance of two different controllers (Wei and Craig, 2009) and to determine the impact of time-of-use electricity tariffs on economic performance (Matthews and Craig, 2013). Although these economic performance functions were not applied to fault conditions in previous work, the approach can be readily translated to assessing the economic impact of faults.

2.2.1. Performance function for PSE

The economic performance functions developed by Wei and Craig (2009) were specifically for the milling circuit modelled by Le Roux et al. (2013), and are considered in this study.

The particle size (measured and simulated as the particle size estimate, PSE) is identified by Wei and Craig (2009) as having the most influence on the economic performance of the milling circuit and downstream processes. A performance function was derived which relates the percent recovery (P) of valuable mineral in the downstream flotation process to the PSE:

$$P = a \times PSE_p^2 + b \times PSE_p + c \quad (5)$$

In Eq. (5), PSE_p is the particle size estimate (in percent), $a = -0.009776$, $b = 1.705$, $c = -2.955$. These parameters are from Wei and Craig (2009), who considered a system which did not operate at optimum recovery (in terms of particle size), as a result of other factors such as environmental impact, electricity costs and throughput limitations. Operating the milling circuit section at the optimum recovery point reduces the overall economic performance of the system (which includes up- and downstream sections). This highlights the importance of selecting economic performance criteria which target overall economic performance, rather than just optimising the performance of smaller sections.

Since one of the simulated faults is a bias in the measurement of particle size, it is important to distinguish between the *actual* (or simulated) and *measured* particle size. The economic performance of the circuit is calculated from the simulated particle size (since this ground truth is known), but the circuit and control system operate according to the measured particle size (even when this is known to be faulty in the simulation). The economic impact of this incorrect operation can be evaluated for the simulated case, but would be difficult for real world data, since the actual particle size would not necessarily be known.

The relationship between the actual (or simulated) particle size estimate (PSE) and valuable metal recovery is shown in Fig. 2; the steady state value for PSE under normal operating conditions is approximately 67% in the current simulation.

The instantaneous value of mineral recovered can be calculated from the throughput, recovery, ore grade and mineral price (Wei and Craig, 2009):

$$\text{Mineral value} = \text{throughput rate} \times \text{head grade} \times \text{recovery} \times \text{mineral price} \quad (6)$$

Wei and Craig (2009) did not consider the impact of throughput changes, but in this case the throughput is represented by the measured variable MFS (mill feed solids, in ton/h). For the purposes of this study, simulating a platinum mine milling circuit, head grade is assumed to be 3 g/ton, and mineral price US\$ 35.3/g (equivalent to US\$ 1000/ounce).

2.2.2. Electricity cost

Matthews and Craig (2013) considered the impact of time-of-use

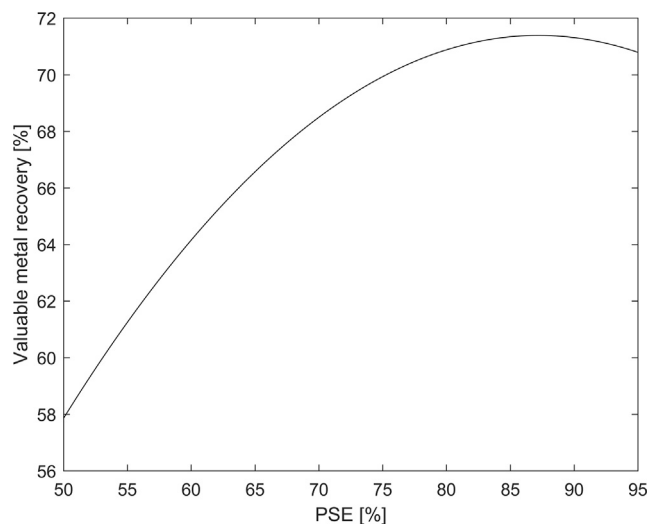


Fig. 2. Metal recovery as a function of particle size estimate (PSE), after Wei and Craig (2009).

electricity tariffs on the operation of a milling circuit, but found that the impact of electricity costs was small compared to the value of mineral recovered. Other studies, for example in the olive oil industry (Báez-González et al., 2016), have found that electricity costs can have a significant impact on process economics, and so for the sake of completeness an estimate of electricity cost due to mill power consumption is included in this study, as it would be applicable to mining operations with lower-value metals than platinum, as is considered here. It may also be important to include the electricity consumption of other equipment, such as pumps or electric furnaces, in other process applications.

The instantaneous electricity cost (in US\$/h) is calculated as the mill power draw (P_{mill} in Table 1, in kW) multiplied by the cost of electricity (assumed to be US\$ 0.06/kWh, the standard tariff in previous studies).

2.2.3. Overall economic performance factor

The economic performance factor (EPF, in US\$/h) of the simulated run of mine milling circuit is defined as the difference between income due to mineral recovery (a function of throughput and recovery) and the cost of electricity to power the mill (a function of the transfer of power from the mill to the mill charge, the feed ore hardness and throughput):

$$\text{EPF} = \text{Mineral value} - \text{Electricity cost} \quad (7)$$

As discussed above, there are other factors which would have an impact on the overall profitability of such an operation, and these should be developed for further implementations of this method.

2.3. Fault detection and diagnosis

Although specific methods for process monitoring are described below, the proposed approach to fault detection and diagnosis is general, and can be readily applied with other methods. Fig. 3 provides a summary of the steps required to implement the statistical monitoring approach, an adaptation of the framework in Aldrich and Auret (2013), which proposes feature extraction from normal operating conditions data in order to monitor new data for faults, and contribution methods to identify variables which may be the cause of detected faults.

2.3.1. Statistical process monitoring

Feature extraction by principal component analysis (PCA) is used as the basis for process monitoring in this work (rather than more complex nonlinear or dynamic variants of PCA, or other feature extraction

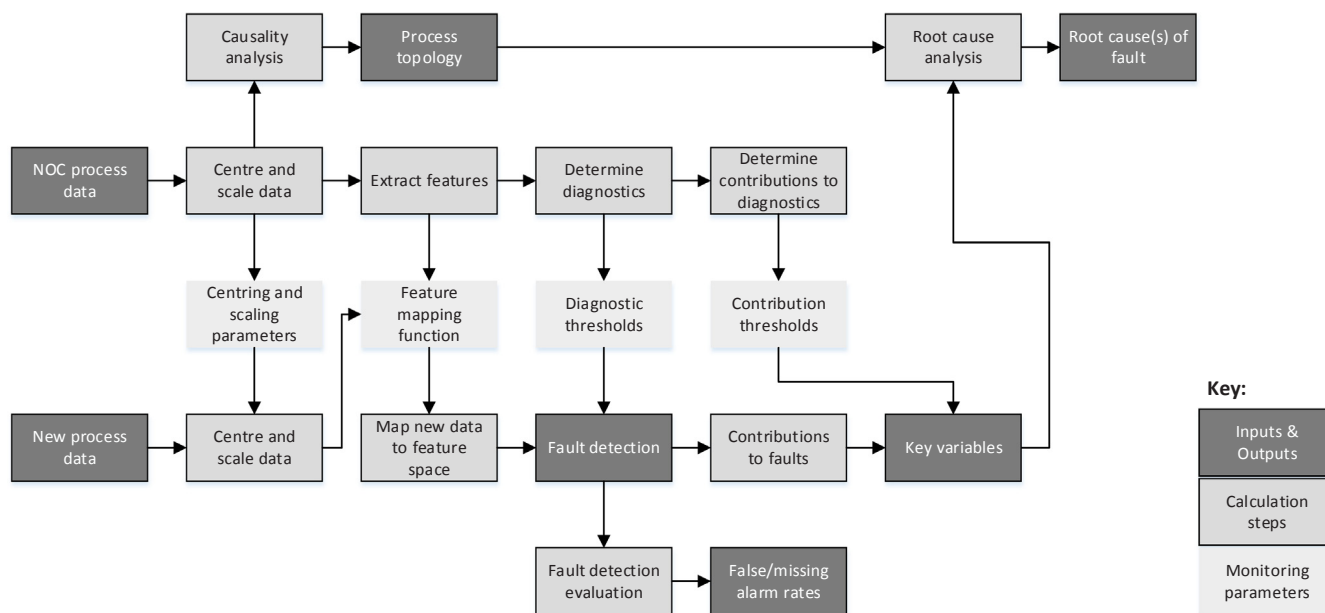


Fig. 3. General statistical monitoring approach.

methods) because it is well described in the literature (Qin, 2012), and is sufficient to demonstrate the benefits of the approach for monitoring of a complex process.

Given a data set, represented by the matrix X (with N rows, the number of samples, and M columns, the number of measurement variables), PCA decomposes the normalised and centred data into the score matrix, T , and the orthogonal vectors (or principal components, represented by the matrix P) which are oriented in the directions of greatest variance in the data set.

$$X = TP^T \quad (8)$$

The columns of P are the eigenvectors of the covariance matrix $X^T X$, and the eigenvalues (λ) of the covariance matrix represent the variance explained by each principal component. In most engineering applications, significant correlation exists between measurement variables, and so the majority of the variance in the data can be captured by the first A principal components, where $A < M$. In this study, A was chosen to retain at least 95% of the total variance, with the remaining variance assumed to be due to noise. This assumption can be checked at implementation through the use of scree plots (Zhu and Ghodsi, 2006) and normality tests such as the quantile-quantile plot. The first A columns of the matrix P , denoted P_A , are the retained principal components. The scores of the PCA model are obtained from the original data (or new data points) as follows:

$$T_A = XP_A \quad (9)$$

The data can be reconstructed in the original measurement space from the scores and principal components. The reconstructed data, $\hat{X}$, is a denoised representation of the original data:

$$\hat{X} = T_A P_A^T \quad (10)$$

Two monitoring diagnostics are derived from the PCA model: the modified Hotelling's T^2 distance and the squared prediction error. Control limits for these diagnostics can be derived analytically, if the data is normally distributed (Qin, 2003), or from historical data. In this paper, the simulated data is subject to large random walk perturbations, relatively minor normally-distributed sensor noise, and control loop interactions; under controlled conditions, the data is expected to remain approximately normally distributed. The NOC training data is used to generate a representative set of values for the diagnostics, and the limits are selected as the 99th percentile of the values. This limit affects the

rate of false and missing alarms for the model, but the selection of the optimum limit is beyond the scope of this work.

A fault is considered to be detected by the monitoring technique when the monitoring diagnostic exceeds the control limit.

The modified Hotelling's T^2 distance for sample i is calculated as follows:

$$(T_A^2)_i = \sum_{k=1}^A \frac{T_{i,k}^2}{\lambda_k} \quad (11)$$

where $(T_A^2)_i$ is the value of the statistic for sample i ; $T_{i,k}^2$ is the square of the k -th column of the score calculated in Eq. (9); and λ_k is the variance explained by the k -th principal component.

The squared prediction error for sample i is determined as follows:

$$SPE_i = \sum_{j=1}^M (X_{ij} - \hat{X}_{ij})^2 \quad (12)$$

where SPE_i is the value of the statistic for the sample i ; M is the number of variables in the process data; X_{ij} is the data point for the j th variable for sample i ; and $\hat{X}_{ij}$ is the reconstructed data point for the j th variable for sample i .

New data is projected into the model (after being normalised and centred) using Eqs. (9) and (10), and the monitoring statistics calculated from Eqs. (11) and (12). A fault is indicated if either monitoring statistic exceeds its threshold.

2.3.2. Contribution plots

Once an alarm has been triggered due to a violation of a diagnostic threshold, the contribution of each of the measured variables to the diagnostic can be determined. There are a number of contribution methods, including complete, partial and reconstruction-based contributions (Alcala and Qin, 2011, 2009). The problem of fault “smearing” can make identification of faults with contribution methods difficult (Garcia and Munaro, 2016; Van den Kerkhof et al., 2013), due to the connections between variables, particularly in complex systems with control loops. For the purposes of this study, complete contributions to the squared prediction error (commonly referred to in the literature simply as “contribution plots”) are used to identify the important variables, as they are simple to apply and sufficient to demonstrate the method.

The contribution $C_{j,i}$ for sample i of variable j to the squared

prediction error (SPE) diagnostic for a true alarm is calculated as the square of the difference between the original data value for sample i of variable j , X_{ij} , and the reconstructed value for sample i of variable j , $\hat{X}_{ij}$:

$$C_{ij} = (X_{ij} - \hat{X}_{ij})^2 \quad (13)$$

The average contribution C_j of variable j is determined as the mean of all $C_{k,j}$ where k indicates all samples for which a true alarm was triggered. This average can be calculated for the entire period of interest, for offline evaluation, or for the period when the fault is first identified, for online evaluation. In this study, the average contribution is determined from the first point exceeding the control limit. A contribution plot can be constructed, displaying C_j for $j = 1, \dots, M$.

For this study, a threshold for the contributions for each variable is calculated from the 99th percentile of the contributions from the NOC training data (Aldrich and Auret, 2013). The ratio of the contributions to the threshold is the relative contribution; a variable with a relative contribution greater than one for a detected fault is considered a key variable for that fault. The key variables may be causal variables (i.e. variables associated with the cause of the abnormal event) or symptom variables (i.e. perturbed extensively due to the abnormal event). The contribution plot does not distinguish causal variables from symptom variables, and so causal relationships between the variables are needed to identify the root causes of faults.

2.3.3. Data-based process topology

Due to the highly interconnected nature of processes, faults may propagate through connected paths in the process and fault symptoms may be seemingly unrelated to the root cause of the fault. A process topology provides information about both connectivity and causality which can be exploited to determine probable propagation paths from the fault symptoms.

A process topology can be derived from fundamental knowledge of the process (Groenewald and Aldrich, 2015; Landman and Jämsä-Jounela, 2016), but this is not always available, or connections between variables are not always understood in complex systems, and so a data-based approach is used here. Data-based causality methods include linear cross-correlation (Bauer and Thornhill, 2008), partial cross-correlation (de la Fuente et al., 2004), transfer entropy (Bauer et al., 2007) and Granger causality (Bressler and Seth, 2011; Granger, 1969). All of these methods identify correlations between pairs of variables in the case where time information is available; it is assumed that when changes in one variable are correlated with changes in another variable at a previous time, then a causal link exists. The second variable is said to cause the first.

Causality methods can be computationally expensive, particularly for systems with large numbers of variables, or with many observations. A reduced set of variables is frequently selected to represent the system, based on process knowledge or monitoring results (Li et al., 2016; Lindner and Auret, 2015; Yuan and Qin, 2014), but this can have the effect of reducing the information which is analysed. The problem of computational expense is becoming less of a concern as available computational power increases, and causality methods become more efficient.

Granger causality is used in this study to develop process topologies, because of its computational efficiency (particularly compared to cross-correlation and transfer entropy methods) and intuitive implementation (Bressler and Seth, 2011; Li et al., 2016; Yuan and Qin, 2014). The implementation is described in the Appendix A.

A question which is not commonly addressed in the literature is the selection of data for the construction of process topologies. The simplest approach, and that which appears to be taken in the literature, is to make use of all available data. This approach requires no prior knowledge of the process, has no hyperparameters to specify, and does not lose any of the information which the causal method may extract

from the data.

However, in the authors' experience, when working with both simulated and operating data, this approach has been unsuccessful. For example, in the current work, application of Granger causality to all of the normal operating conditions (NOC) data produced process topologies in which every variable appeared to have a causal link with every other variable. This result is neither useful nor sensible, and so two other proposals are presented here.

These proposed methods are based on the application of Granger causality to windows of data from NOC. An obvious drawback is that this requires the specification of window width, which in turn requires knowledge of the dynamics of the process of interest. Here, the window width is selected based on a multiple of an order of magnitude estimate of a representative residence time from the milling circuit. An order of magnitude estimate is used to reduce the requirement for expert system knowledge; in this case, the estimate was 30 min, a reasonable value for a large industrial operation (for comparison, the simulated mill has a steady state residence time of approximately 15 min). The multiple of the residence time to determine the window width was 100, which ensured that the windows were each large enough to include sufficient variability in the data for the Granger causality application. The effect of these assumptions is further discussed in the results.

The first proposed method applies Granger causality to every possible consecutive (but non-overlapping) window within the NOC data. The significance of each causal connection is averaged over all of the windows, and used to construct the process topology. This approach is attractive as it makes use of all of the NOC data, but at a reduced computational cost, and produces fewer spurious connections (as will be shown in the results).

The second proposal is to apply Granger causality to a single window of data which ends at the time of fault detection. This method is attractive as it has a very small computational cost, and considers a portion of data similar to the faulty data, potentially improving fault diagnosis.

Once a fault is detected, the key variables (from the contribution method) are identified, and the causal variables determined from the process topology by tracing backwards (in terms of causality).

One of the limitations of data-based root cause analysis is apparent when considering the final step. The nodes in a topology map represent measured variables in the process, but faults are not always directly associated with a specific sensor measurement. For example, pump impeller wear is not measured directly, but may have an impact on measured variables such as pressure or flow. Therefore the root cause analysis can, at best, point to a variable that is associated with the root cause of the fault.

3. Results and discussion

3.1. Dynamic mill circuit simulation with process disturbances and abnormal events

The results of the simulation of the run of mine milling circuit are shown below for a period of 134 days: 90 days of normal operating conditions (NOC) data, to train the statistical monitoring model; 14 days of NOC data to test the performance of the monitoring method; and 30 days of fault data, for each of the two faults (particle size estimate fault, and mill liner fault).

The data to generate the initial steady state of the model is not shown.

3.1.1. Process inputs and outputs

The process inputs and outputs for the particle size estimate (PSE) fault are shown in Figs. 4 and 5; the process inputs and outputs for the mill liner fault are shown in Figs. 6 and 7.

The main source of variation in the NOC periods of data is due to the incorporation of random walks in the feed ore size distribution

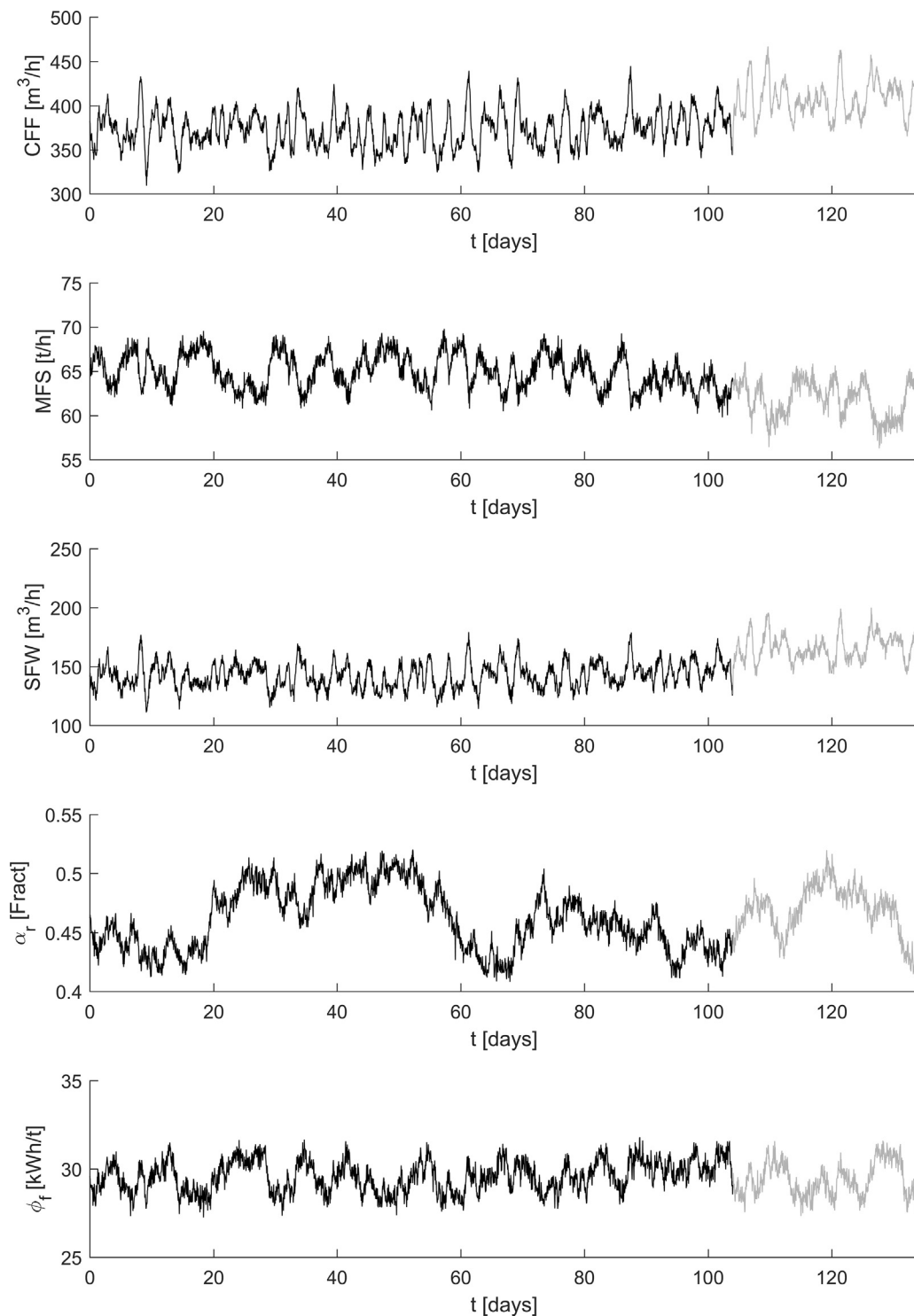


Fig. 4. Process inputs for the PSE fault simulation (data subsampled at 1 h intervals). NOC data shown in black, and fault data in grey.

variation (defined as the fraction of rocks in the feed ore, α_f) and feed ore hardness (defined as the power needed per tonne of fines produced, Φ_f). These variations ensure a robust model of normal operating conditions, which includes the responses of various control loops to a number of disturbances.

Note that the data in Figs. 4–7 has been subsampled from the simulation resolution of 30 s to 1 h, for clarity of the figures; all calculations with the data were at full available resolution unless stated otherwise.

3.1.2. Economic performance factors

The economic performance factors for each of the simulated cases are shown in Figs. 8 and 9. These figures include the predicted economic performance which would have resulted in the absence of the simulated faults. The reader is reminded that the fault-free data from the simulation (the “ground truth”) would not normally be available for real operating data, and is not used by the process monitoring system in this work. The fault-free data is presented to illustrate the degradation in economic performance which occurs as a result of the faulty conditions which are simulated, and to emphasise the potential economic

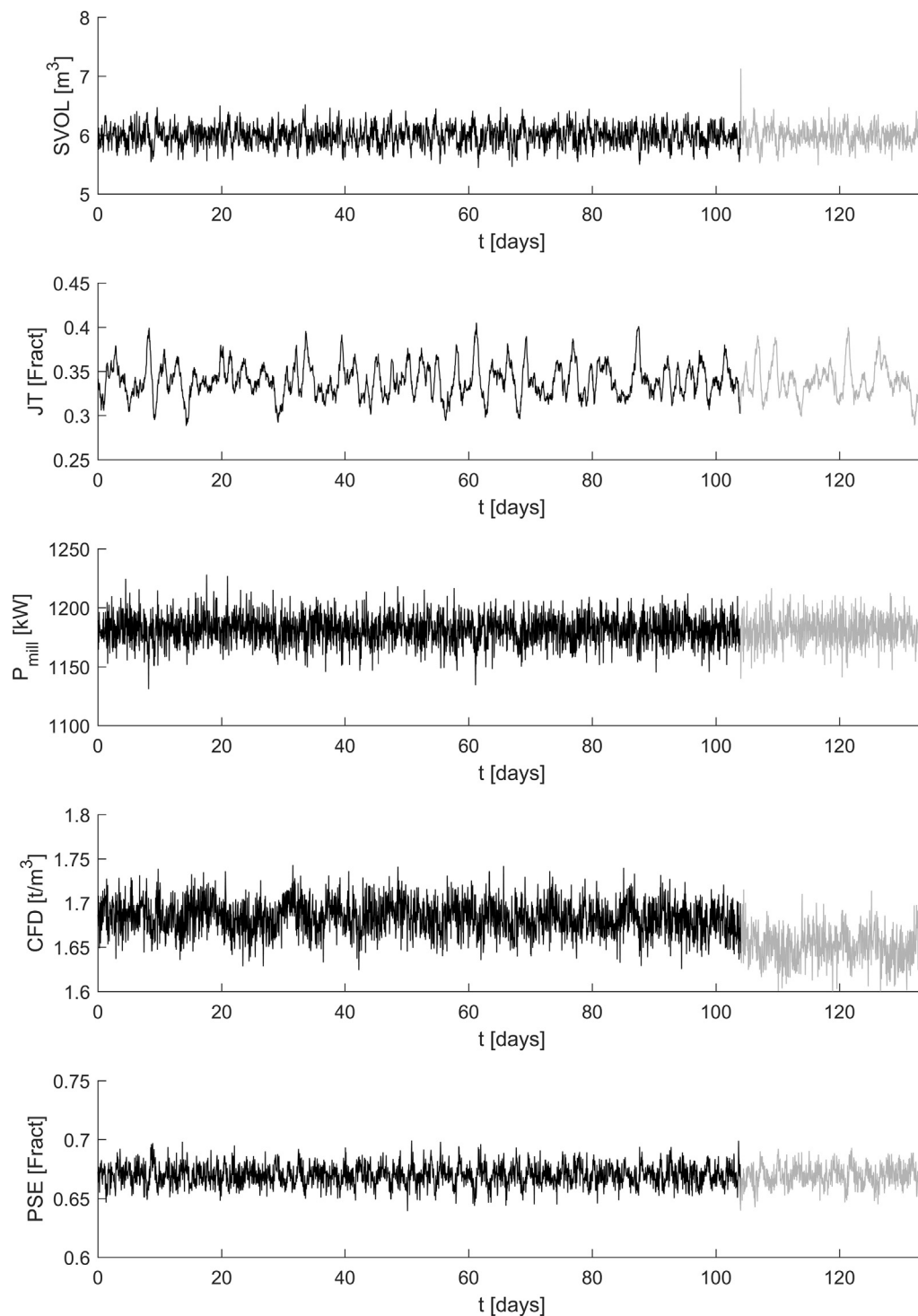


Fig. 5. Process outputs for PSE fault simulation (data subsampled at 1 h intervals). NOC data shown in black, and fault data in grey.

impact of fault detection and diagnosis.

Fig. 8 shows the economic performance factor for the PSE fault. There is a clear decrease in economic performance after the fault occurs (the dark grey line in Fig. 2), when compared to the performance that would have been achieved in a fault-free condition (the light grey line in Fig. 2). Even though the particle size is actually smaller than before the fault (resulting in higher recovery, from the relationship in Fig. 2), the throughput of the system decreases due to a larger circulating load in the recycle, which is a result of the faulty particle size sensor. The decrease in throughput outweighs the increase in recovery.

The economic performance of the simulated milling circuit is not

abnormally low following the PSE fault (when compared to the NOC data), and monitoring of the plant based only on economic performance would not necessarily identify the performance shown in Fig. 8 as abnormal. As will be shown below, however, statistical process monitoring techniques can detect and identify faults which can cause reduced economic performance.

Fig. 9 shows the economic performance of the milling circuit for the mill liner wear fault simulation. Because this is a gradual fault, a steady decline in economic performance is apparent (particularly when compared to the fault-free economic performance). This is as a result of the decrease in throughput, a direct result of the decreasing efficiency of

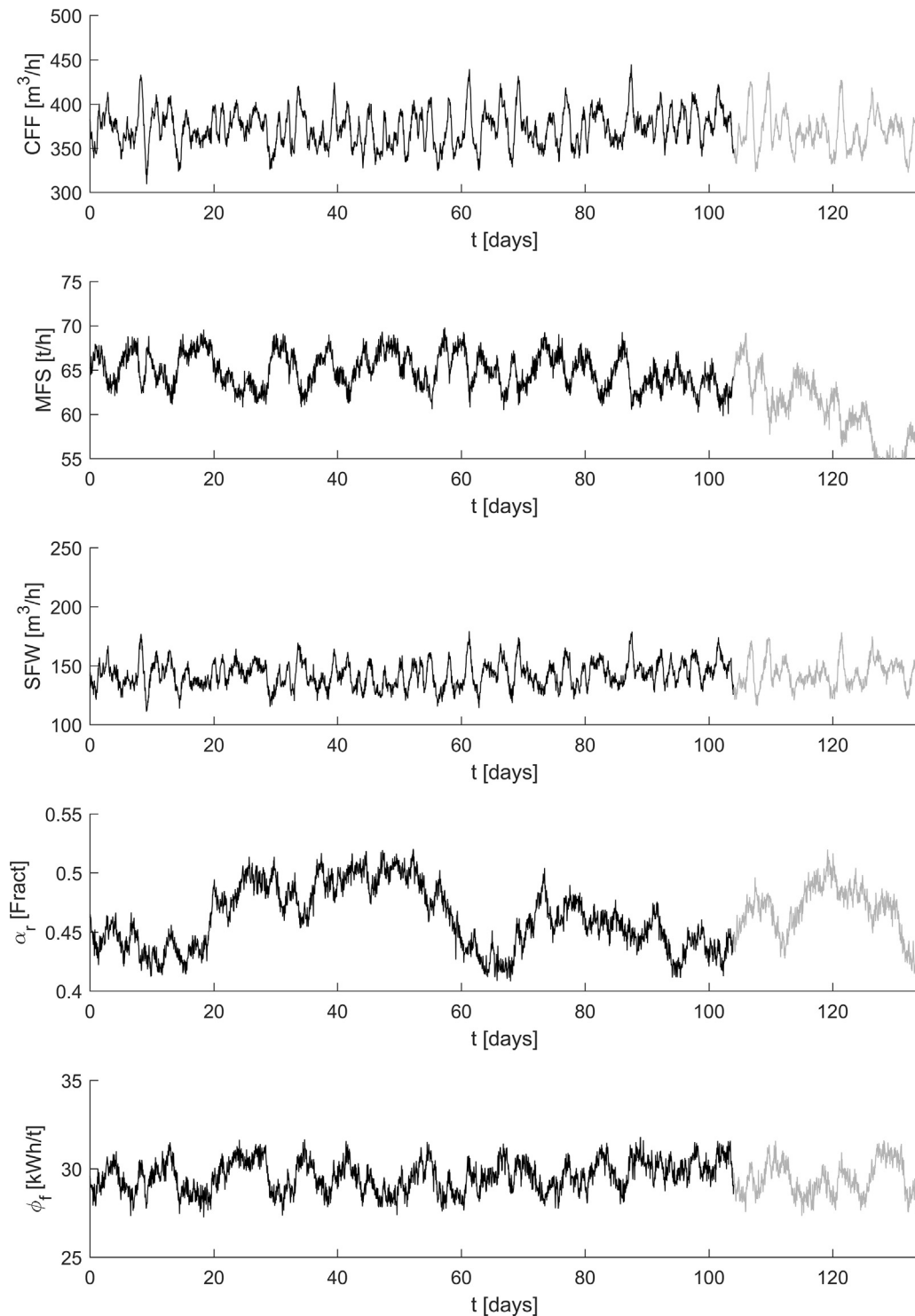


Fig. 6. Process inputs for mill liner wear fault simulation (data subsampled at 1 h intervals). NOC data shown in black, and fault data in grey.

the energy transfer from the mill to the mill charge.

The decreasing performance as a result of the mill liner wear would become apparent from monitoring economic performance as those shown in Fig. 9, but only after the trend had continued for an extended period; it is difficult to distinguish, for example, between the decreasing economic performance observed during normal operation from day 20 to day 30 (a result of fluctuations in the inputs to the system) and the decreases observed in the first 20 days after the fault (a result of input fluctuations and degrading mill liner performance). Statistical monitoring can detect this type of faulty condition.

3.2. Statistical process monitoring for fault detection

The statistical monitoring model was trained on the normal operating conditions data, making use of 6 retained principal components, which explained > 98% of the variance in the data, as shown in the scree plot in Fig. 10. The score vectors of the two non-retained principal components were shown to be normally-distributed (see Figs. 23 and 24 in the Appendix A), validating the assumption that these components represented random variation such as sensor noise.

The results of statistical monitoring of the simulated milling circuit

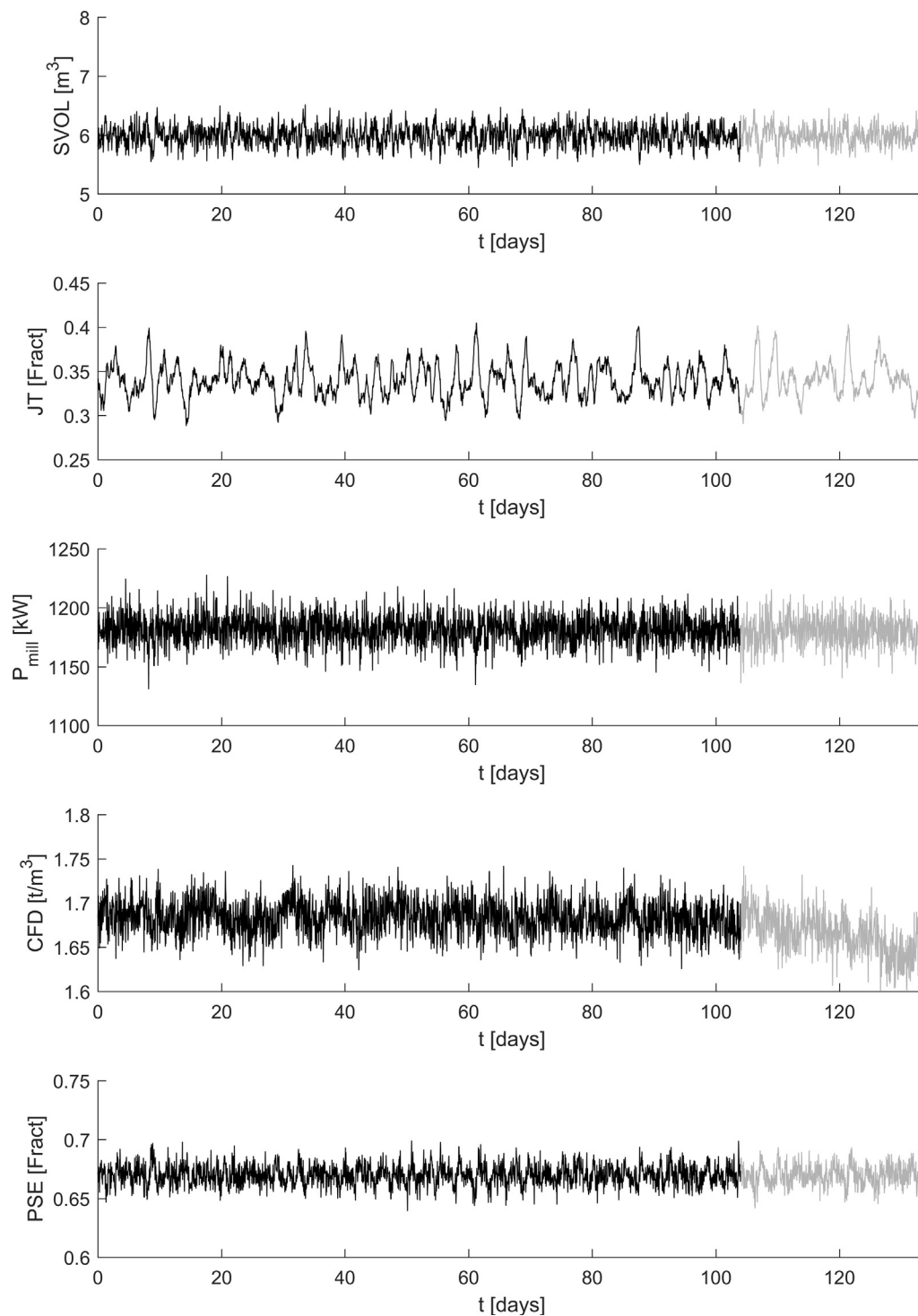


Fig. 7. Process outputs for mill liner wear fault simulation (data subsampled at 1 h intervals). NOC data shown in black, and fault data in grey.

are shown below, for each of the simulated faults.

3.2.1. PSE fault

The control charts for each of the monitoring statistics (modified Hotelling's T_A^2 , and squared prediction error, SPE) are shown in Figs. 11 and 12, and the detection performance in terms of false, missed and true alarm rates and detection delay at the specified detection limits is shown in Table 3.

For both monitoring statistics, the false alarm rate (determined from the NOC test data) is low, reflecting successful training of the model on the NOC training data.

The missing alarm rate (determined from the fault test data) is particularly high for the modified Hotelling's T_A^2 ; although the statistic detects the fault in the initial stages of the fault simulation, the statistic does not detect the fault once the control system has brought operation back within control. The SPE monitoring statistic displays much better detection of this fault during the fault test data period, since the faulty sensor reading reflects a change in the correlation structure of the data. This type of fault is best detected by the SPE statistic.

In all cases, the detection delay reflects very rapid detection of the faulty condition.

Fig. 13 shows the economic performance for fourteen days before

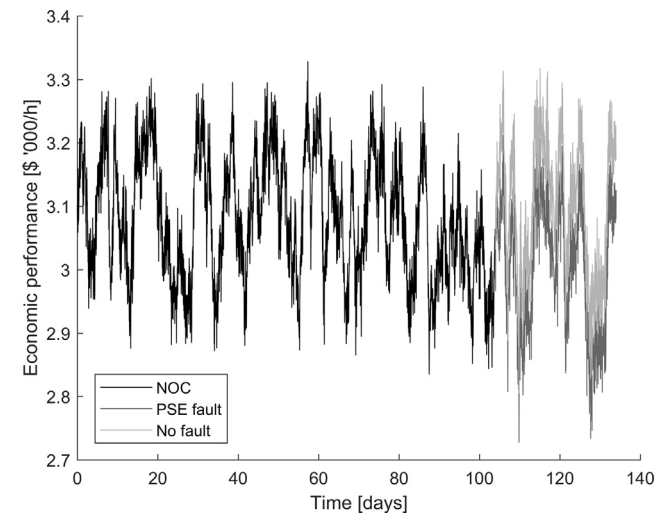


Fig. 8. Economic performance for PSE fault simulation. NOC data shown in black, fault data in dark grey, and fault-free data in light grey.

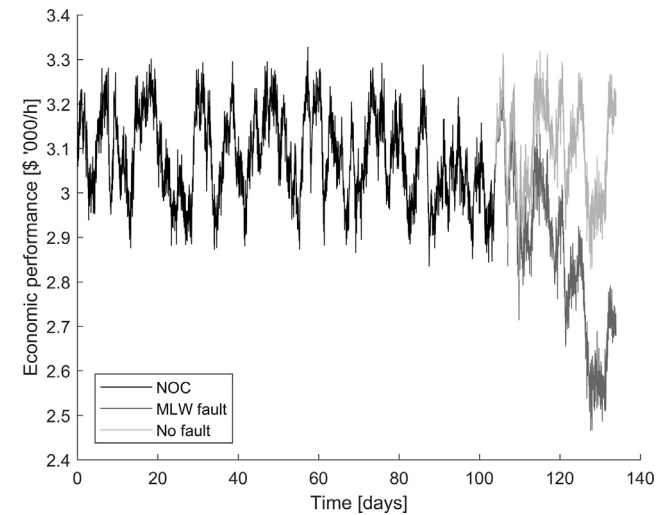


Fig. 9. Economic performance for mill liner fault simulation. NOC data shown in black, fault data in dark grey, and fault-free data in light grey.

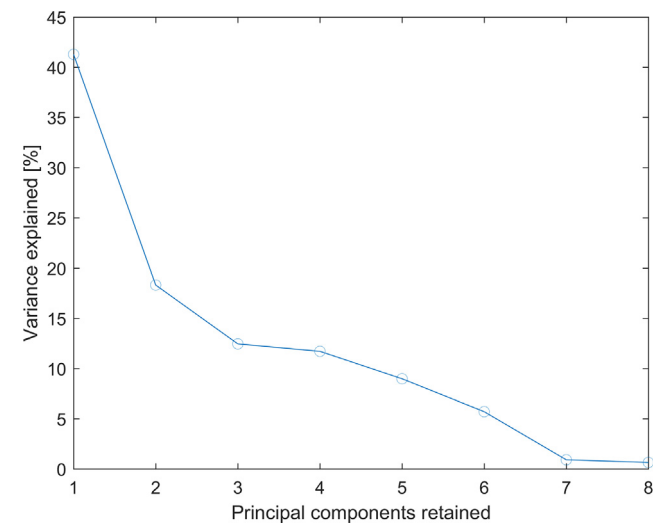


Fig. 10. Scree plot of variance explained per principal component in monitoring model.

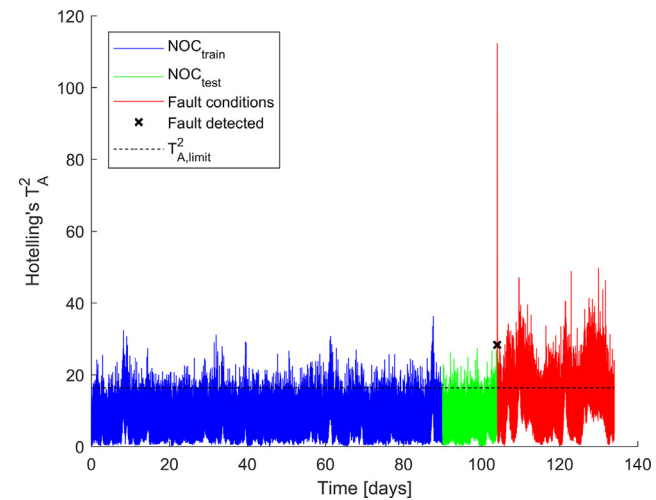


Fig. 11. Control chart for modified Hotelling's T_A^2 , for PSE fault.

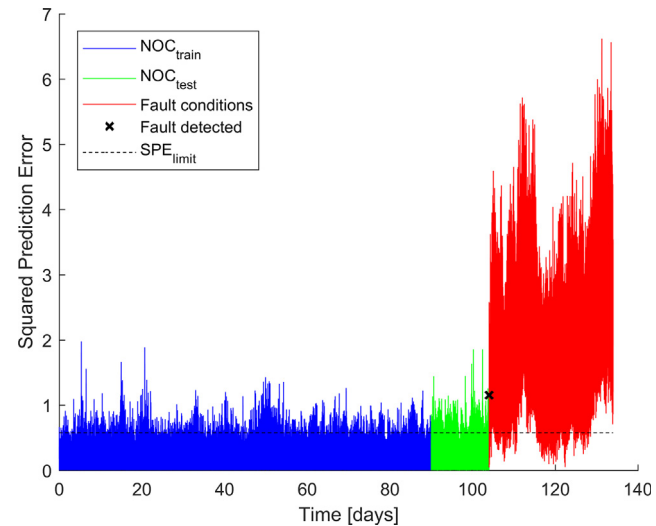


Fig. 12. Control chart for squared prediction error, for PSE fault.

Table 3
Detection performance for PSE fault.

	False alarm rate [%]	Missed alarm rate [%]	True alarm rate [%]	Detection delay [h]
Hotelling's T_A^2	0.595	78.4	21.6	0.00
SPE	2.24	1.48	98.5	0.0830
Both T_A^2 and SPE	2.83	1.19	98.8	0.0830

and after the fault began, at 104 days, and indicates the first detection of the faulty condition. Fault detection and diagnosis (such as by the methods described in this paper) may allow operators to correct the faulty condition, and avoid the period of decreased economic performance (indicated by the difference between the fault and fault-free data).

3.2.2. Mill liner fault

The control charts for the two monitoring statistics are shown in Figs. 14 and 15, and the detection performance for the selected thresholds is shown in Table 4. Once again, the false alarm rates are low, reflecting successful training of the model on the NOC data.

The poorer performance of the statistics in terms of missed alarms (compared with the PSE fault) is a result of the nature of the fault: a

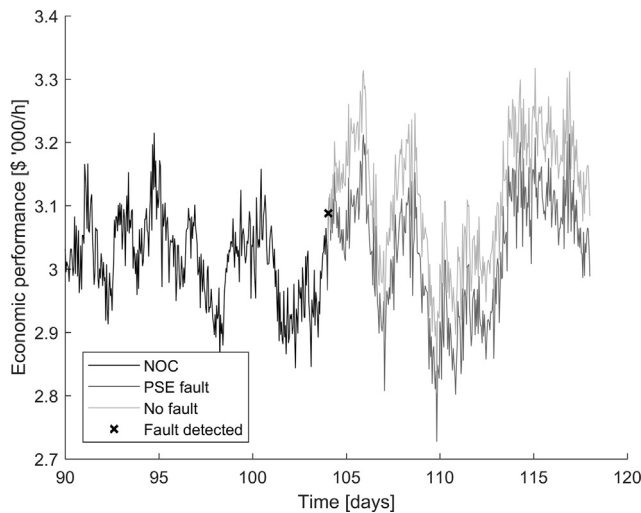


Fig. 13. Economic performance for PSE fault simulation. NOC data shown in black, fault data in dark grey, and fault-free data in light grey. The time of first fault detection is shown as a black X.

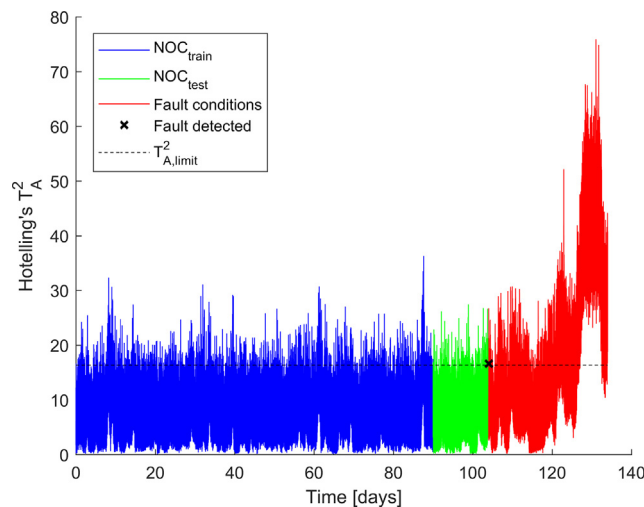


Fig. 14. Control chart for modified Hotelling's T_A^2 , for mill liner fault.

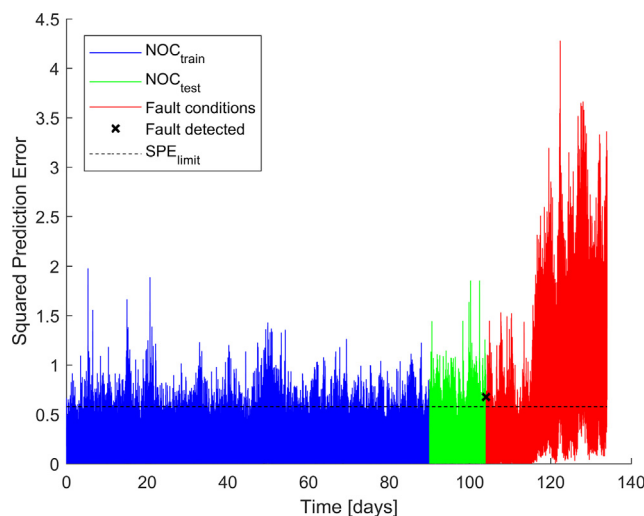


Fig. 15. Control chart for squared prediction error, for mill liner fault.

Table 4

Detection performance for mill liner fault.

	False alarm rate [%]	Missed alarm rate [%]	True alarm rate [%]	Detection delay [h]
Hotelling's T^2	0.595	64.1	36.0	0.783
SPE	2.24	50.4	49.6	0.183
Both T^2 and SPE	2.83	44.5	55.5	67.5

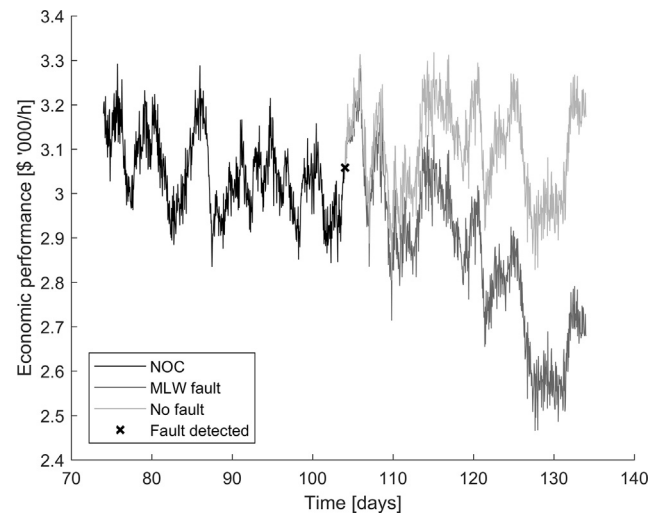


Fig. 16. Economic performance for mill liner fault simulation. NOC data shown in black, fault data in dark grey, and fault-free data in light grey. The time of first fault detection is shown as a black X.

gradual change in the mill liner condition is difficult to detect, and the slow occurrence means the control system maintains control of the milling circuit, resulting in the poor performance of the T_A^2 statistic and the delay in detection by the SPE statistic. Detection of incipient faults is an ongoing challenge in the field of fault detection and identification (Ji et al., 2016; Ruiz-Cárcel et al., 2016; Yang and Xiao, 2012).

Due to the incipient nature of this fault, the detection delay is longer than for the PSE fault; simultaneous monitoring does not detect the fault for more than two months, for example. However, the detection delays for the individual monitoring statistics are significantly shorter than would be possible if only the economic performance (shown in Fig. 9) were monitored.

Fig. 16 shows the economic performance for 30 days before and after the fault started, and indicates the point at which statistical monitoring detected the faulty condition. The subsequent large decrease in performance could be avoided following fault detection and diagnosis. This early detection is one of the significant benefits of multivariate monitoring over the monitoring of single parameters, such as the economic performance factor: the fault can be detected by multivariate monitoring well before the decrease in economic performance becomes apparent.

3.3. Fault diagnosis

As shown in Figs. 13 and 16, faults can be detected relatively quickly, and there is an economic benefit to diagnosing the faults, and fixing the underlying problem. The diagnosis of the detected faults by contribution and causality methods is described below. The discussion highlights some of the challenges of implementation, and the need for further development of the methods.

3.3.1. Contribution plots

The relative contributions to the SPE statistic at the time of detection for the PSE fault are shown in Fig. 17; the identified key variables

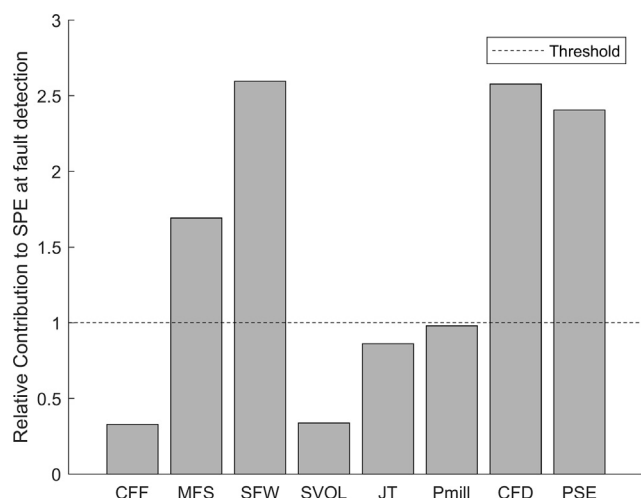


Fig. 17. Relative contribution plot for PSE fault.

are the mill feed solids (MFS), sump feed water (SFW), cyclone feed density (CFD) and particle size estimate (PSE). The contribution method has reduced the number of variables of interest from eight to four, which can be used for further diagnosis, such as using the process topology.

Fig. 18 shows the relative contributions of each of the measured variables to the SPE statistic for the mill liner fault at the time of detection. For this fault, all of the variables except the cyclone feed flow rate (CFF), sump feed water (SFW) and cyclone feed density (CFD) are identified as key variables; the contribution method has not significantly reduced the number of variables of interest. This is due to the gradual nature of the fault; as the fault extent increases, the entire circuit responds, due to the interactions which result from the recycle and control loops. As mentioned above, the challenge of detecting gradual faults is an ongoing area of research in the field.

3.3.2. Data-based process topology

As mentioned in the Methods section, although the application of Granger causality appears to be simple, requiring the user to specify only the significance level for retained connections and the maximum time lag, the authors' experience has shown that this is not the case. In particular, generating the process topology using all available NOC data resulted in an over-connected network, in which every variable appeared to have a causal relationship with every other variable.

For this reason, two alternative approaches were proposed here: to

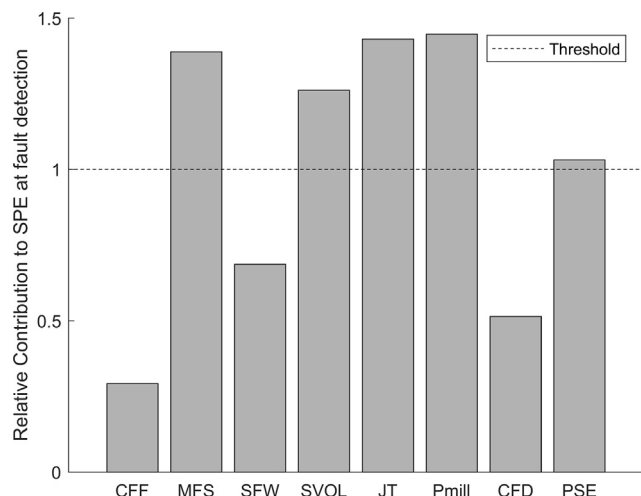


Fig. 18. Relative contribution plot for mill liner fault.

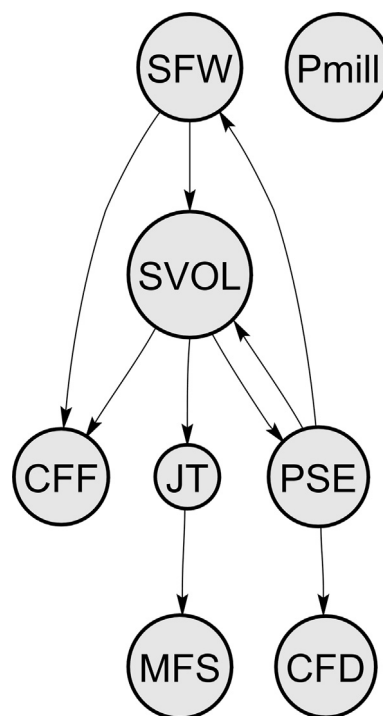


Fig. 19. Process topology generated using Granger causality applied to consecutive windows in NOC data. Level of significance 0.01. Model lag 5.4 min (determined by Akaike Information Criterion).

define a window size (in terms of a representative process lag or residence time for the process), and then generate the process topology from consecutive windows of data from the NOC data, and to generate the topology from a single window of data which ends at the time of fault detection. The order of magnitude estimate for the representative process lag was selected as 30 min for this study, and the window size was specified as 100 times this process lag value. The results of the topology generation were found to be relatively insensitive to the choices of these values, suggesting that either approach may be suitable.

The process topology generated for consecutive windows in the NOC data is shown in Fig. 19, and the topology generated for a single window ending at the fault detection is shown in Fig. 20. Both topologies were generated at a significance level of 0.01 for causal connections. The connectivity could also be visualised in the form of a cause-effect matrix, particularly for larger systems, in which the directed graph method may become overly complex.

Both of the process topologies reflect the control loop links between the particle size estimate (PSE) and sump feed water (SFW), between the sump volume (SVOL) and the cyclone feed flow rate (CFF), and between the mill feed solids (MFS) and the mill charge (JT). However, other connections are less intuitive: for example, the mill power (P_{mill}) appears to cause changes in the mill feed solids (MFS) in Fig. 20, and is not causally connected to any of the variables in Fig. 19.

The variability of the causal connections between process variables was investigated in further detail (as described in the Appendix A). The causality method appears to be sensitive to the data period selected for analysis, even for this simulated case. The random walks in the feed ore particle size and hardness, which were implemented to mimic actual variation of this input variable, and the simulated sensor noise in all measured variables, lead to variation in the response of the milling circuit, as well as the control loops. Depending on the behaviour of the simulated system within each window of data, the causality method produces different process topologies. Application of the method to real operating data (rather than simulated data) would be even more challenging, as additional issues such as missing data, variable sampling

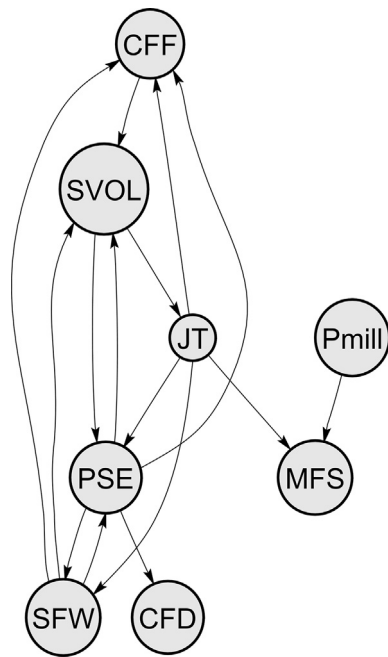


Fig. 20. Process topology generated using Granger causality applied to window ending at fault detection. Level of significance 0.01. Model lag 4.5 min (determined by Akaike Information Criterion).

rates and operator interventions would further obfuscate the causal links between process variables. This result suggests that further research is required in the field to formalise the method's application to real-world situations.

3.3.3. Root cause analysis

The results of the root cause analysis for the PSE fault are shown in Fig. 21; the variables of interest are highlighted in blue, and the

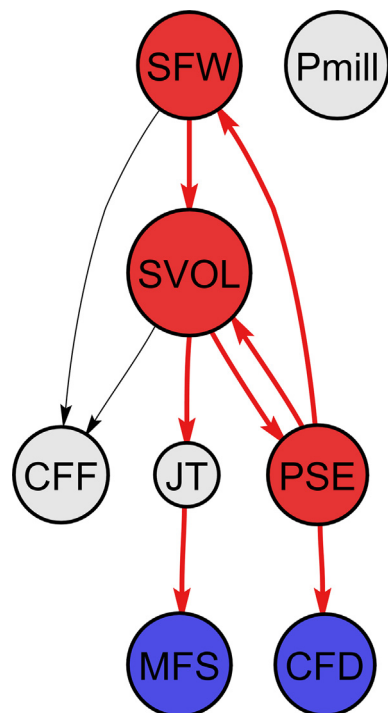


Fig. 21. Root cause analysis for PSE fault. Process topology generated using Granger causality applied to consecutive windows in NOC data. Level of significance 0.01. Model order 5.4 min (determined by Akaike Information Criterion).

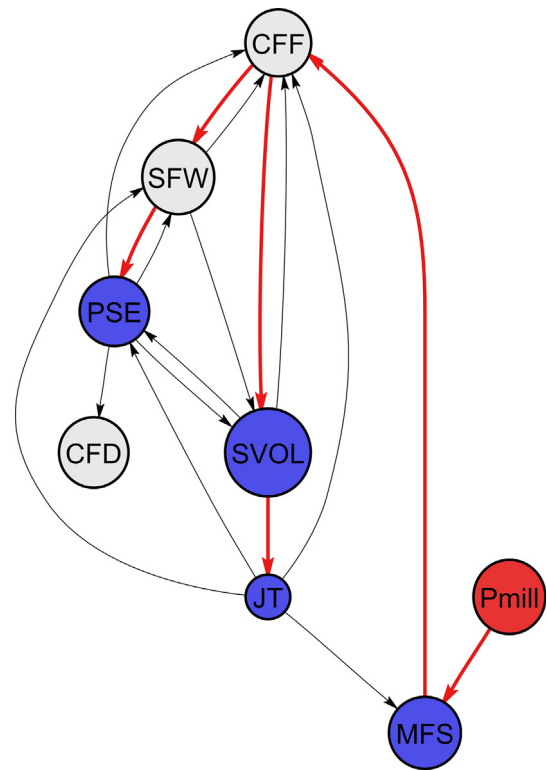


Fig. 22. Root cause analysis for mill liner fault. Process topology generated using Granger causality applied to window ending at fault detection. Level of significance 0.05. Model order 4.5 min (determined by Akaike Information Criterion).

variables which are identified by backpropagation as potential causes are highlighted in red. Propagation paths are also shown in red. The topology was generated from the consecutive windows in the NOC data, as the results of root cause analysis with the window ending at fault detection were inconclusive.

The analysis suggests that the root causes of this fault are the sump feed water (SFW), sump volume (SVOL) and particle size estimate (PSE). Reducing the number of variables would enable plant support personnel to more closely study the operation of the hydrocyclone; the control loop link between the particle size and sump feed water (both root causes) may assist the identification of the particle size sensor as the real root cause.

The results of the root cause analysis for the mill liner fault are shown in Fig. 22; the variables of interest (as identified in the contribution plot, Fig. 18) are highlighted in blue, and the potential root cause nodes in red. Propagation paths are shown in red. Note that this topology was generated from the window of data ending at the fault detection, at a significance level of 0.05, as no root causes could be identified at a significance level of 0.01, or using the topology generated from the consecutive windows approach.

The analysis identifies the mill power (P_{mill}) as the root cause of the fault. As mentioned previously, one of the limitations of a data-based root cause approach is that, at best, it can identify variables associated with a fault, if the actual fault is not directly associated with a measured variable. In this case, the identification of the mill power as a causal variable may be sufficient to allow the diagnosis of the actual fault with the mill liner.

4. Conclusions

This paper described a statistical monitoring approach based on feature extraction (by principal component analysis) and fault detection (with statistical thresholds), followed by fault diagnosis using contribution methods and data-based process topologies (using Granger

causality). The monitoring approach was illustrated using standard techniques, but can be easily extended to more complex methods.

A simulation of the run of mine milling circuit described by [le Roux et al. \(2013\)](#) was developed to investigate the applicability of fault detection and root cause analysis algorithms for industrial use. The milling circuit consisted of a semi-autogenous mill, sump and hydro-cyclone with recycle from the cyclone underflow to the mill. Simple control loops were implemented in the simulation. Process fluctuations were simulated by using random walks, to reflect the non-deterministic but temporal nature of changes to the inputs to such a milling circuit, and sensor noise was included. An economic performance function for the milling circuit was developed, in order to assess the impact of the faulty conditions, and the potential benefit of implementing fault detection and identification.

Two fault conditions were simulated: a faulty particle size analyser on the product stream (PSE fault) and rapid wear of the mill liner (mill liner fault). The mill circuit simulation and simulated fault data have been made available for further research, such as testing other fault detection and diagnosis methods, or investigating the economic impact of fault detection in more detail.

The statistical monitoring method detected the sudden PSE fault almost instantaneously, and the contribution and process topology analysis suggested that the root causes of the fault were the sump feed water (SFW), sump volume (SVOL) and particle size estimate (PSE). Reducing the number of variables of interest to these three would allow plant support personnel to more rapidly identify the actual root cause. The PSE fault caused a decrease in economic performance for the milling circuit, which could be avoided or drastically reduced with the implementation of statistical monitoring methods as described here.

The statistical monitoring method took slightly longer to detect the mill liner fault, due to the incipient nature of the fault. This detection is, however, significantly faster than could have been achieved by monitoring only the economic performance of the circuit, for example. The contribution and process topology analysis identified the mill power (P_{mill}) as the root cause of the fault. The close association between the

mill power and the mill liner itself suggests that the analysis would be useful in diagnosing the fault. Correct diagnosis of the fault would reduce the degradation in economic performance which occurred as a result of the mill liner wear.

The results of the process topology generation and root cause analyses have shown that Granger causality has the potential to identify variables associated with the cause of faults in complex systems, such as the milling circuit simulated in this study. The sensitivity of the method to the data used to generate process topologies was highlighted in this work, suggesting a need to further formalise the method's application, particularly to large datasets. Two approaches for the selection of data for topology generation were proposed, making use of simple assumptions about the process, in order to avoid the need for expert knowledge.

Supplementary material

The simulation of the milling circuit, as well as the data used in this paper, have been made available for further use. The code can be downloaded from <https://github.com/ProcessMonitoringStellenboschUniversity/ME-milling-circuit>. Instructions for the use of the simulation and data files are included with the source code. The user requires a copy of MATLAB and Simulink.

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Appendix A

A.1. Granger causality implementation

[Granger \(1969\)](#) developed a formalisation of Wiener's causality description in the context of regression models, called Granger causality: when inclusion of past measurements of X in a regression model of Y results in an improvement on the prediction of the autoregressive model of Y , then X is said to have a causal influence on Y . Consider the time series of two variables, $X(t)$ and $Y(t)$. $Y(t)$ can be modelled as a univariate autoregressive (AR) model, known as the restricted model, as shown in Eq. (14).

$$Y(t) = \sum_{j=1}^k B_j Y(t-j) + \epsilon_y(t) \quad (14)$$

In this case only past values of Y are incorporated to predict future values of itself. k is the model order defining the time lag; B is the AR coefficient; and ϵ_y is the prediction error. $Y(t)$ can also be modelled incorporating past values $X(t)$ and $Y(t)$, using multivariate AR models, known as the full model, as shown in Eq. (15).

$$Y(t) = \sum_{j=1}^k [A_{yx,j} X(t-j) + A_{yy,j} Y(t-j)] + \epsilon_{y|x}(t) \quad (15)$$

A_{yx} and A_{yy} represent the model coefficients. When the variance of $\epsilon_{y|x}(t)$ is less than the variance of $\epsilon_y(t)$, it implies that the prediction of Y is improved by including past values of X . X is then said to Granger cause Y . The Granger causality can be quantified as shown in Eq. (16).

$$F_{x \rightarrow y} = \ln \frac{\text{var}(\epsilon_y)}{\text{var}(\epsilon_{y|x})} \quad (16)$$

When the prediction of Y is not improved by including X , $\text{var}(\epsilon_{y|x}) = \text{var}(\epsilon_y)$ and $F_{x \rightarrow y} = 0$. When the prediction of Y is improved by including X , $\text{var}(\epsilon_{y|x}) < \text{var}(\epsilon_y)$ and $F_{x \rightarrow y} > 0$. A_{yx} , A_{yy} and B can be determined using the least squares approach.

This pairwise approach to Granger causality has been shown to predict spurious causal links, and a multivariate extension of Granger causality proposed ([Li et al., 2016](#)); the multivariate extension was applied in this study. The full model, shown in Eq. (15), is extended to the prediction of the data matrix X (with N rows, the number of samples, and M columns, the number of measurement variables), using previous values of X , as shown in Eq. (17).

$$X(t) = \sum_{i=1}^M \sum_{j=1}^k [A_{ij} X_i(t-j)] + \epsilon_{full}(t) \quad (17)$$

A_{ij} represents the autoregressive model coefficient for the i th variable of X at the j th time lag, $X_i(t-j)$ represents the j th lagged value from the i th column of the matrix X , and ϵ_{full} represents the prediction error of the model. The restricted model is calculated by excluding variable m from Eq. (17), as shown in Eq. (18).

$$X(t) = \sum_{i=1, i \neq m}^M \sum_{j=1}^k [A_{ij} X_i(t-j)] + \epsilon_{restricted}(t) \quad (18)$$

Although the method is based on the fitting of linear autoregressive models to data, it is anticipated that a nonlinear process under effective control will remain in an approximately linear regime. The method is thus applicable to normal operating conditions data for a process under control. As the process deviates from normal conditions, the resulting nonlinearity may result in approximation errors when constructing process topologies, and so fault data is not used for this purpose in this paper.

The model order, k , can be determined using the Akaike Information Criterion (AIC). For a model of M variables, the AIC is given as shown in Eq. (19) (Akaike, 1973).

$$AIC(k) = \ln(\det(\Sigma)) + \frac{2kM^2}{N} \quad (19)$$

where Σ is the residual covariance matrix of the full model and N is the number of observations. The AIC is evaluated for a range of model orders, and the model order giving the lowest AIC is chosen.

Time series generated by processes will include some stochasticity, which means that nonzero values for Granger causality will be calculated even when there is no connection between the variables. Hypothesis tests to determine the statistical significance of the values obtained need to be employed. For Granger causality this hypothesis test can take the form of an F-statistical test (Bressler and Seth, 2011; Yuan and Qin, 2014), as shown in Eq. (20).

$$\frac{\frac{RSS_r - RSS_f}{k}}{\frac{RSS_f}{N - 2k - 1}} \sim F_{k, N-2k-1} \quad (20)$$

$$RSS_r = \sum_{t=k+1}^N \epsilon_y^2(t) \quad (21)$$

$$RSS_f = \sum_{t=k+1}^N \epsilon_{y|x}^2(t) \quad (22)$$

N represents the sample size of the time series, RSS_r and RSS_f represent the sum of squares of the residuals in the restricted and full models respectively. An F-test can then be performed at a chosen significance level, α , which is usually 0.01 or 0.05 (Yuan and Qin, 2014).

A.2. Statistical monitoring results

Figs. 23 and 24 show the results of normality tests for the non-retained principal components' score vectors. The close agreement between the quantiles of the score vectors and the quantiles of a standard normal distribution validates the assumption that the non-retained principal components represent random variation in the data, and can be excluded from the statistical monitoring model.

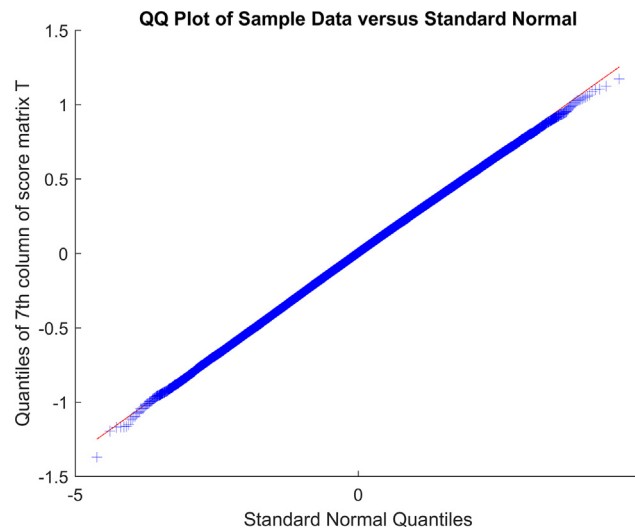


Fig. 23. Quantile-quantile plot of the score vector for the seventh (non-retained) principal component.

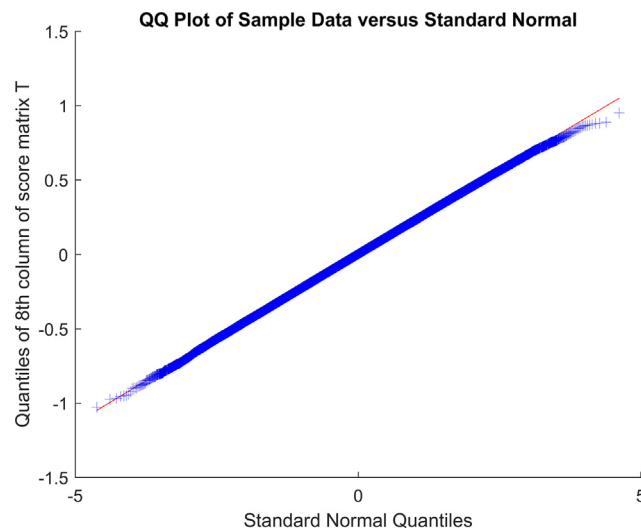


Fig. 24. Quantile-quantile plot of the score vector for the eighth (non-retained) principal component.

A.3. Process topology results

In order to investigate the variability of the process topologies produced by the Granger causality method, the significance of the causal connections between process variables was plotted for each of the windows in the normal operating conditions (NOC) data, which was used to construct the process topology shown in Fig. 19. The figures below show histograms of the upper tail probabilities from the F-test for selected variable pairs.

Figs. 25–27 show the results of the connections between the manipulated and controlled variables for the three control loops described in Table 2. In all three cases, the controlled variable is shown to have a causal relationship with the manipulated variable (for a significance level of 0.01). This result suggests that Granger causality can reliably detect strong causal connections (such as those in control loops).

However, the method does not always detect expected connections, such as between the mill feed solids (MFS) and mill power (P_{mill}), as shown in Fig. 28. At a significance level of 0.01, a causal connection between these variables was detected in less than 5% of the windows in the NOC data.

Similarly, the method detects unexpected causal connections between process variables. For example, as shown in Fig. 29, there is a significant causal connection from the sump volume (SVOL) to the mill charge (JT) in all data windows (at a significance level of 0.01). Although interactions between variables are expected, due to the recycle and control loops, it is highly counterintuitive that the sump volume could cause changes in the mill charge as identified by Granger causality.

These results highlight the sensitivity of Granger causality to the portion of data used to construct the process topology: not only are expected connections not always (or even frequently) identified, unexpected and counterintuitive connections can be identified.

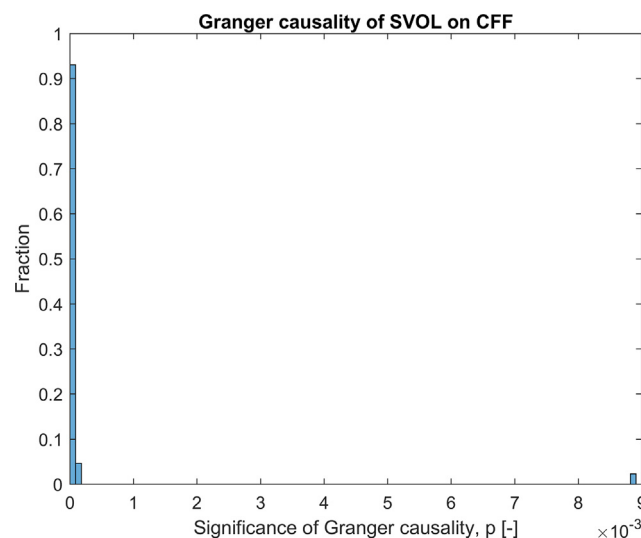


Fig. 25. Granger causality of the control loop pairing between sump volume (SVOL, controlled variable) and cyclone feed flow rate (CFF, manipulated variable).

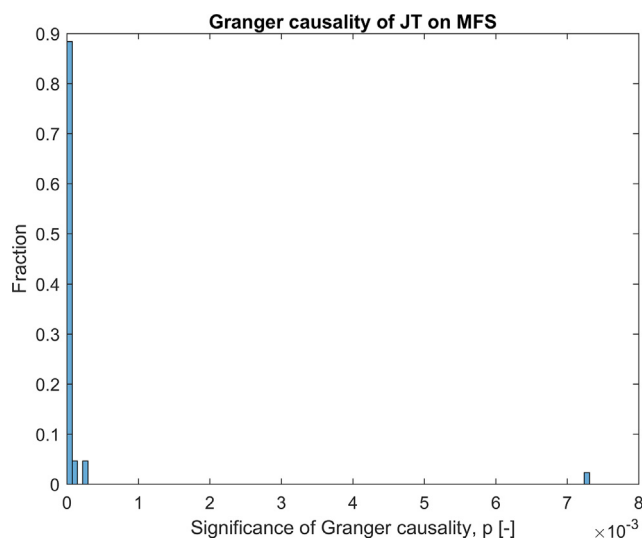


Fig. 26. Granger causality for the control loop pairing between mill charge (JT, controlled variable) and mill feed solids (MFS, manipulated variable).

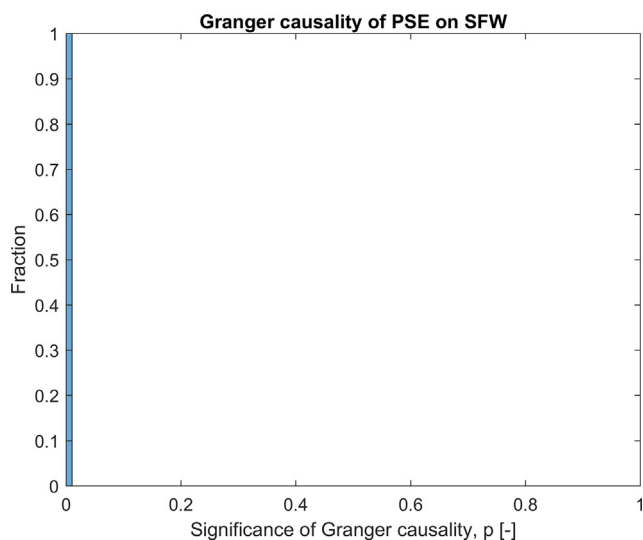


Fig. 27. Granger causality for the control loop pairing between particle size estimate (PSE, controlled variable) and sump feed water (SFW, manipulated variable).

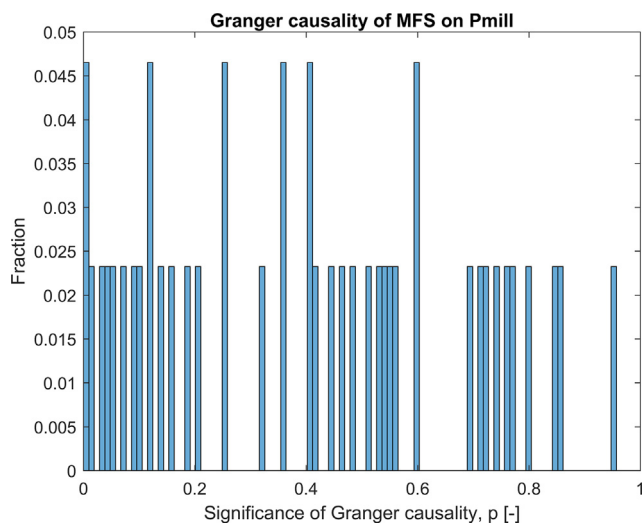


Fig. 28. Granger causality of an expected pairing - large variability.

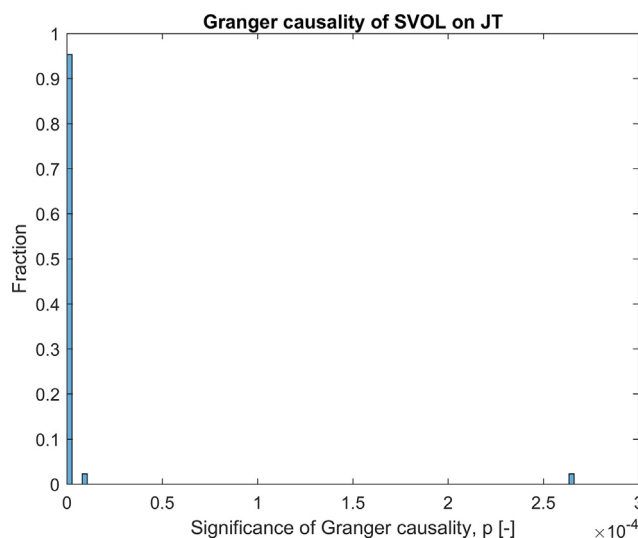


Fig. 29. Granger causality of an unexpected pairing – significant.

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